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STATE OF THE ART REVIEW
Epidemiology, prevention, and health care policies

Pulmonary embolism epidemiology: trends, risk factors, comorbidities, and clinical implications

Marco Zuin [1,2,3,*,†], Cecilia Becattini [4,†], Frederikus A. Klok[5,6], Behnood Bikdeli [7,8,9], Stefano Barco [6,10], Ioannis Farmakis[6], Roman Chopard[11,12,13], Irene M. Lang[14], Claudio Bilato [15], Elvira Grandone[16,17,18], Samuel Z. Goldhaber[8], and Gregory Piazza [7,8]

[1]Department of Translational Medicine, University of Ferrara, Via Luigi Borsari 46, Ferrara 44100, Italy; [2]Department of Cardio-Thoraco-Vascular Sciences and Public Health, University of Padova, Padua 35128, Italy; [3]Department of Cardiology, Madre Teresa di Calcutta Hospital, AULSS 6, South Padova Hospitals, Schiavonia, Italy; [4]Internal, Vascular and Emergency Medicine—Stroke Unit, University of Perugia, Perugia, Italy; [5]Department of Internal and Vascular Medicine, University Medical Center Leiden, Leiden, the Netherlands; [6]Center for Thrombosis and Hemostasis, University Medical Center of the Johannes Gutenberg University Mainz, Mainz, Germany; [7]Division of Cardiovascular Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, USA; [8]Thrombosis Research Group, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, USA; [9]YNHH/Yale Center for Outcomes Research and Evaluation, New Haven, CT, USA; [10]Department of Angiology, University Hospital Zurich, Zurich, Switzerland; [11]Department of Cardiology, Centre Hospitalier Universitaire de Besançon, Besançon, France; [12]SINERGIES Laboratory, University Marie and Louis Pasteur, Besançon, France; [13]F-CRIN, INNOVTE Network, Saint-Étienne, France; [14]Department of Cardiology, Medical University of Vienna, Vienna, Austria; [15]Department of Cardiology, West Vicenza Hospitals, Arzignano, Italy; [16]THROMBOSIS and Hemostasis, Fondazione Casa Sollievo della Sofferenza IRCCS, San Giovanni Rotondo, Italy; [17]B/GYN, University of Foggia, Foggia, Italy; and [18]OB/GYN, I M Sechenov First Moscow State Medical University, Moscow, Russian Federation

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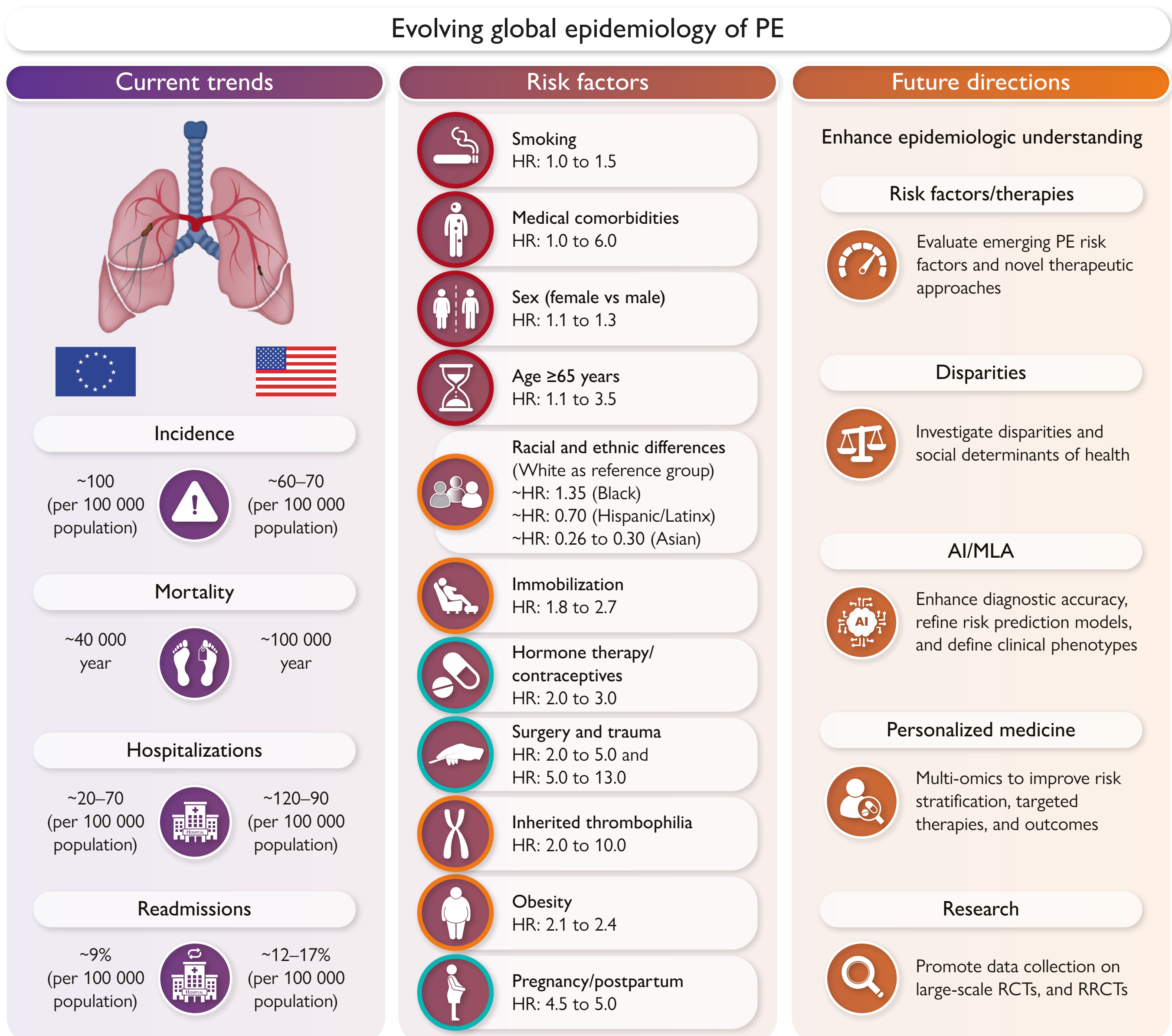
* Corresponding author. Email: marco.zuin@edu.unife.it
† Equally contribute.

Graphical Abstract



Current trends in the epidemiology, understanding of risk factors, and future directions for evaluation and management of PE. The left panel illustrates the contemporary burden of PE in Europe and the United States, including incidence, hospitalizations, and readmission rates. The central panel summarizes major clinical risk factors with corresponding HR, such as obesity, immobilization, pregnancy/puerperium, hormone therapy, surgery, trauma, smoking, medical comorbidities, aging, sex differences, inherited thrombophilia, and racial/ethnic disparities. Risk factors are categorized as persistent (red circle), transient (green circle), or both persistent and transient (orange circle). The right panel highlights emerging priorities aimed at enhancing epidemiologic understanding, including evaluation of novel risk factors and therapies, reduction of disparities, integration of AI/ML-based tools, personalized medicine, and advancement of clinical research

AI, artificial intelligence; **HR**, hazard ratios; **MLA**, machine learning algorithms; **PE**, pulmonary embolism; **RCT**, randomized controlled trial; **RRCT**, registry-based randomized controlled trial

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Abstract

As the global population ages and individuals live longer with chronic diseases associated with venous thromboembolism, acute pulmonary embolism (PE) is expected to remain a major public health challenge. Like myocardial infarction and stroke, PE is linked to established cardiovascular risk factors, including advancing age, obesity, smoking, and chronic inflammatory conditions. Despite this, population-based primary and secondary prevention strategies for PE remain limited, highlighting the need for an updated epidemiological understanding. A comprehensive public health approach to PE should encompass not only the management of acute events and transient risk factors but also a detailed appreciation of epidemiology and risk patterns across populations and communities, to support clinician education, public awareness, long-term individual and community risk assessment, ultimately preventive efforts. In this review, we summarize current epidemiological evidence, highlighting trends on modifiable and non-modifiable risk factors for acute PE, with the goal of informing strategies for improved prevention and population health management.

Keywords Pulmonary embolism • Epidemiology • Risk factors • Comorbidities • Prevention

Introduction

Due to its wide spectrum of clinical presentations and potential for rapid deterioration and death, pulmonary embolism (PE) continues to challenge even the most seasoned clinicians.[1] Cancer, a common condition in an ageing population, has become an increasingly important risk factor for PE, especially since such patients are living longer due to therapeutic advances.[2] Additionally, the growing prevalence of risk factors such as obesity and heart failure (HF) has further influenced the epidemiology of PE.[3] The rising incidence and prevalence of PE may reflect not only a global increase in both transient and persistent risk factors[4] but also greater use of advanced imaging, such as CT pulmonary angiography (CTPA) along with heightened awareness of the disease.[5] Given these evolving factors, the present review aimed to illustrate the current epidemiology of acute PE to better inform effort to improve population outcomes.

Methodology

This narrative review summarizes English-language studies from the past 20 years on acute PE epidemiology, risk factors, and population-specific data. Studies were selected by consensus among the co-lead authors (MZ and CB) and the senior author (GP), in discussion with coauthors. A comprehensive search of PubMed/Medline, Embase, and Scopus, along with reference screening (see Supplementary data online, *Table S1*), ensured inclusion of all relevant studies. Disagreements were resolved through web-based discussion and consensus. The search was last updated on 29 November 2025.

Overall epidemiology

The incidence and prevalence of PE are rising worldwide, driven by population ageing and the growing burden of cardiovascular risk factors.[1] Expanded use of CTPA and greater clinical and public awareness have increased detection, including subclinical cases. True PE incidence likely remains underestimated due to nonspecific symptoms and misdiagnosis.

Venous thromboembolism (VTE) affects an estimated 10 million people globally each year.[6] In Europe, countries such as France, Italy, Germany, Spain, Sweden, and the United Kingdom (UK) collectively report over 700 000 cases. In the United States, 1 million cases are diagnosed annually.[7,8] The incidence rates vary significantly, ranging from 66 to 104 cases per 100 000 in the United States and Europe to 6.7 per 100 000 in Japan, likely due to myriad genetic, environmental, and diagnostic differences.[9]

In the United States, approximately 60 000 to 100 000 deaths are attributed to acute PE each year, with some estimates suggesting the number may exceed 100 000.[10] In Europe, PE causes approximately 39 000 deaths annually.[11] However, when considering the broader category of VTE, which includes both PE and deep vein thrombosis (DVT), annual mortality rises to an estimated 544 000 deaths.[10–12] Additionally, in the United States, PE incidence is increasing, with age-adjusted rates rising 2.1% annually among adults 25–64 years.[13] Major risk factors include obesity, diabetes, smoking, sedentary lifestyle, cancer, surgery, and cardiovascular disease, contributing to ∼40% of VTE-related hospitalizations.[14] In Western Europe, incidence reaches 104 per 100 000, largely due to population ageing and improved diagnostic approaches. Healthcare-associated VTE remains a major issue, accounting for roughly 60% of events during or after hospitalization for medical or surgical illness.[15]

Mortality

Short-term mortality rate for acute PE remains noteworthy. Among untreated patients with haemodynamically unstable PE, up to 30% die within 30 days, and 11% may succumb within the first hour of hospital presentation.[16] In contrast, subsegmental PE, currently accounting for approximately 15% of acute PE diagnoses,[17] is associated with mortality rates typically <1%.[17] In the United States, between 1999 to 2019, high-risk PE was listed as the underlying cause of death in 209 642 patients, corresponding to an age-adjusted mortality rate (AAMR) of 3.01 per 100 000 people (95% CI: 2.99 to 3.02).[18] Moreover, PE-attributable mortality increased by 1.5% annually among young adults aged between 25–44 years during this period.[18] Although Europe's population is more than double that of the United States, PE-attributable deaths are higher in the United States, reflecting greater prevalence of key risk factors such as obesity and cardiovascular disease, disparities in healthcare access, regional variations in emergency care, and differences in coding and reporting practices.

Geographic variability in PE mortality is evident. In China, the PE-attributable mortality was reported as 1.0 per 100 000 population in 2021, substantially lower than in Western countries,[19] perhaps paralleling lower reported incidence.[19] Conversely, in Japan, recent data suggest rising mortality possibly reflecting demographic shifts and lifestyle westernization.[20]

Despite advances in diagnostic and therapeutic strategies, mortality in high-risk PE remains considerable. In-hospital mortality ranges from 20% to 47% and may approach 85% in cases complicated by cardiac arrest.[21] PE-related sudden cardiac death in the United States has increased and disproportionately impacts men, White and Hispanic/Latinx individuals, adults <65 years, rural communities, and resident of the Southern states, likely highlighting persistent demographic and regional disparities.[22] Meanwhile, increased detection of low-risk and incidental PE has contributed to an apparent decline in overall mortality, partially reflecting a dilution effect rather than solely due to improvements in treatment outcomes, as emphasized in registries such as the COntemporary management of PE (COPE).[23] This phenomenon underscores the importance of distinguishing true improvements in outcomes due to therapeutic advances from shifts in diagnostic thresholds.

Long-term complications

Post-PE syndrome (PPES), defined as persistent or worsening symptoms, functional limitations, or objective cardiopulmonary impairment lasting ≥6 months after acute PE, is common, though its true incidence remains uncertain due to variable definitions.[24] However, the true incidence of PPES remains uncertain due to the absence of a standard definition across studies and the struggle to distinguish whether symptoms are sequelae of the PE or due to other comorbidities.[24] Up to 40%–60% of PE survivors may experience some forms of PPES, with severe manifestation in up to 6% of cases.[25] The FOCUS study was a prospective cohort study of more than 1000 patients with a dedicated structured follow-up for up to two years after the acute event and showed that nearly 16% of patients suffered from post-PE impairment, defined as a combination of persistent or worsening clinical, functional, biochemical, and imaging parameters during follow-up.[26] Exercise intolerance is frequent: greater than one-third of patients reporting dyspnoea and reduced 6-min walk distance at 3 months and ∼20% with persistence at 1 year with ∼15% demonstrating severe limitations on cardiopulmonary exercise testing.[26]

Chronic thromboembolic pulmonary hypertension (CTEPH), a debilitating cause of progressive right ventricular (RV) failure, affects 0.22–0.64 per 100 000 globally and is often underrecognized, partly because some patients lack a clear PE history despite imaging evidence of chronic thromboembolic disease.[27] Reported CTEPH incidence after PE varies widely (2.3%–8.2%), reflecting differences in screening and population characteristics.[25,28] Prospective studies with strict diagnostic criteria show a lower 2-year incidence (∼0.8%), whereas a meta-analysis estimated a pooled incidence of ∼2.7% over 6 months to nearly 9 years of follow-up.[29]

Hospitalization and readmission

Hospitalization patterns for PE show rising incidence but shorter stays. In the United States, adjusted PE hospitalization rates increased from 120 to 187 per 100 000 beneficiary-years between 1999 and 2015, while length of stay fell from 7.7 to 5 days.[30] National Inpatient Sample data similarly show admissions climbing from 23 to 65 per 100 000 between 1993 and 2012, with median stay decreasing from 8 to 4 days.[31] Comparable trends are seen in Europe: in Spain, PE incidence rose from 20.5 to 35.9 per 100 000 (2001–18) with length of stay decreasing from 11 to 7 days while in Germany, discharge diagnoses increased from 55.3 to 71.7 per 100 000 (2000–06).[32] Despite higher diagnostic incidence, the absolute number of hospitalizations declined in many Western countries, largely due to widespread CTPA utilization, identifying more isolated or subsegmental, low-risk PE cases suitable for outpatient management.[33] Many of these cases involve patients at low risk of adverse outcomes and may be safely managed in the outpatient setting.[34] The PE-specific risk-adjusted readmission frequency decreased modestly from 1.2% to 1% in the 30-day cohort and from 1.4% to 1.2% in the 90-day cohort from 2010 to 2018.[35] In a *post hoc* analysis of the YEARS study, PE-associated unscheduled readmissions were 9.7% for patients treated at home and 8.6% for initially hospitalized patients.[36] Overall, 30-day readmission after acute PE is estimated at ∼12%–17%.[35,37]

Pulmonary embolism risk factors

Although classifying VTE as 'provoked' or 'unprovoked' is widely used, it oversimplifies recurrence risk by ignoring persistent or combined risk factors.[38] Current literature instead recommends categorizing risk factors as transient or enduring, and further as major or minor. This framework better supports prevention strategies: addressing modifiable transient factors can reduce situational PE risk, while recognizing enduring factors should guide long-term risk mitigation, particularly in high-risk populations (*Figure 1* and *Table 1*).

Risk factors and populations at risk

Age

Advancing age is a major PE risk factor, linked to reduced mobility, endothelial dysfunction, accumulating comorbidities, and a shift towards a prothrombotic state.[39] For individuals <65 years, the annual incidence of PE is estimated at approximately 1–2 cases per 1000 person-years.[40,41] However, this rate increases dramatically in those aged 65–85 years, reaching around 5–6 cases per 1000 person-years.[42,43] Among individuals >85 years, the incidence can exceed 10 cases per 1000 person-years.[44] Of all patients hospitalized with PE, approximately 50.5% are categorized as elderly (>65 years of age), and of those, 28% would be considered to fulfil frailty criteria.[45]

Sex

The epidemiology of acute PE demonstrates significant sex-based trends, despite overall similar age-adjusted incidence rates between women and men. Women exhibit higher PE rates during early and mid-adulthood, whereas men demonstrate increased incidence after 60 years.[46] Among individuals >70 years of age, PE is more prevalent in women (accounting for 58% of cases in this age range).[46,47] Moreover, the risk nearly doubles for both sexes in patients >75 years compared with younger individuals.[13,48] Comorbidities influencing the severity and the outcome of PE also vary by sex. Women have higher rates of hypertensive heart disease, diabetes, and obesity, while men exhibit greater prevalence of chronic obstructive pulmonary disease (COPD) and cancer.[49,50] Regarding mortality, men face a greater risk of sudden death from PE.[22] Moreover, recent analyses suggest no significant sex-informed disparities in PE-related mortality trends among cancer patients or young adults aged 25–44 years in the US.[51,52]

Racial and ethnic differences

Racial and ethnic disparities impact the epidemiology of acute PE. The incidence of PE is inherently higher among certain ethnoracial groups.[53] Black Americans have roughly twice the risk of PE

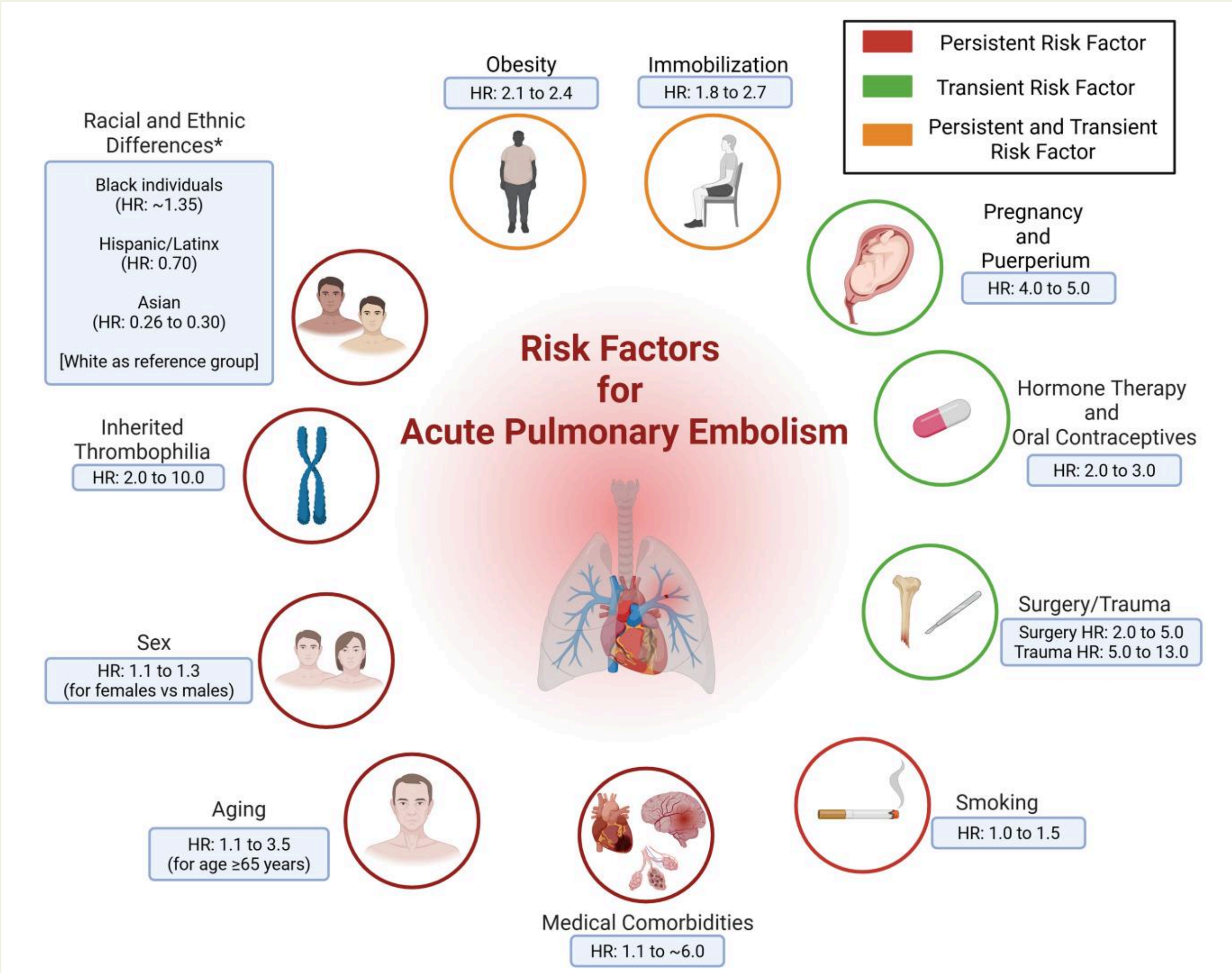

Figure 1 Transient and persistent risk factors and modifiers for acute pulmonary embolism. All risks are expressed as hazard ratios to enable comparability across different diseases. The values shown represent the minimum and maximum hazard ratios identified from the literature search. HR, hazard ratio. *Also a risk modifier

hospitalization and higher mortality than their White counterparts, often presenting with more severe disease at a younger age.[54,55] Despite similar in-hospital mortality rates in some studies, adjusted analyses demonstrate that Black Americans have 30% greater odds of 30-day mortality compared to White subjects when controlling for disease severity.[55,56]

Hispanic/Latinx patients are under-represented in existing healthcare disparities research on PE, despite comprising a significant portion of the US population.[57] Hispanic/Latinx patients are less likely to present with high-risk PE, with only 19% presenting with high-risk cases compared to 25% in non-Hispanic/non-Latinx patients. Additionally, they have lower rates of PE-specific interventions (15% vs 19%) but similar inpatient mortality rates (6.8% vs 7.5%) compared to non-Hispanic/non-Latinx patients.[57] Notably, Hispanic/Latinx individuals appear to have a lower incidence of PE mortality despite facing similar healthcare challenges as other socioeconomically-disadvantaged populations.[58]

Pregnancy and puerperium

Pregnancy and puerperium increase the risk of VTE, with a 4–10-fold higher risk compared to non-pregnant individuals.[59] Pregnancy-related VTE occurs in ~1–2 per 1000 pregnancies, with PE accounting for 20%–30% of cases.[60,61] Risk rises throughout gestation with a peak in the postpartum period: incidence reaches ~160 per 100 000 woman-years in the first two weeks post-delivery, ~20-fold higher than during pregnancy.[62] Other data suggest that the risk of VTE may extend to 12 weeks postpartum.[63] Caesarean delivery further increases risk and contributes to most fatal postpartum PE cases.[64] Regional variations exist, with studies reporting 0.45–0.93 PE cases per 1000 births in Europe and the US.[64] Key risk factors include inherited and acquired thrombophilia, advanced maternal age, history of VTE, obesity, use of assisted conception including *in vitro fertilization*, and pregnancy complications such as miscarriage, preeclampsia or postpartum haemorrhage.[65] Despite prophylaxis advances, pregnancy-associated PE mortality remains notable (1.96 deaths

Table 1 Risk Factors for Acute Pulmonary Embolism

Category	Risk factor/modifier	Clinical insights and mechanisms
Demographic factors	Advanced age	• Incidence increases markedly with age • >10/1000 person-years in those >85 years due to immobility • Endothelial dysfunction • Comorbid burden
	Sex differences	• Women: higher risk during reproductive years • Men: increased incidence and mortality post-60 years • Sex-specific comorbidity profiles (e.g. HF, cancer, COPD)
	Family history	• First degree- family history
	Race and ethnicity	• Black individuals have 2× higher hospitalization and increased 30-day mortality • Hispanic/Latinx patients underrepresented in data, with variable presentation rates
Reproductive and hormonal	pregnancy/puerperium	• 4–10× risk increase (especially urgent caesarean section) • Postpartum period particularly high-risk (up to 159.7/100 000) • Risk amplified with caesarean section and assisted reproductive therapies
	Hormonal therapy and oral contraceptives	• Oestrogen-containing therapies (especially oral) significantly increase PE risk • Highest in the first year of use and with third- and fourth-generation progestins
Infectious triggers	Acute infections	• Promote a transient prothrombotic state via inflammation; • Endothelial injury and immobilization • COVID-19 increases PE risk up to 8 months post-infection.
Chronic cardiovascular disease	Heart failure	• Up to 3× higher mortality in PE patients with HFrEF • Promotes venous stasis and systemic inflammation • SGLT2i and incretin-focused therapies may reduce risk indirectly.
	Coronary artery disease	• Recent myocardial infarction increases PE risk 6-fold • Shared mechanisms include platelet activation and endothelial dysfunction.
	Peripheral artery disease	• Peripheral artery disease independently associated with PE (HR: 1.12) • Abnormal ABI predicts greater risk • Risk of PE directly increases with the number of vascular beds involved.
	Atrial fibrillation	• Both a risk factor and consequence of PE • New-onset atrial fibrillation post-PE linked to worse outcomes and elevated long-term mortality
Chronic pulmonary and renal	Chronic lung disease (COPD, asthma)	• COPD exacerbations may mask PE symptoms • Increased short-term mortality • Asthma associated with 2–3× increased risk
	Chronic kidney disease (CKD)	• Stage 3–4 CKD confers a 2.1× VTE risk • Bleeding risk complicates anticoagulation management
Oncologic and inflammatory	Active cancer	• High PE burden (10%–35%) • Incidental PE common • Immunotherapy and metastasis increase risk; high recurrence rates
	Chronic inflammatory disease	• RA, SLE, IBD significantly increase PE risk • Flares and glucocorticoids amplify thrombotic risk
	Chronic liver disease	• Cirrhosis and MASH associated with increased PE risk (OR: 1.65–4.33) • Coagulation abnormalities play key role
Lifestyle-related factors	Obesity	• 2.2× PE risk • Linked to venous stasis, inflammation, and altered coagulation • Underrepresented in PE severity scores despite strong prognostic relevance • Diet including fruit and vegetables reduces the risk of PE as well as foods with high anti-inflammatory potential.
	Sedentary behaviour	• Independent PE risk factor • 2–3× increased risk in sedentary individuals • Physical activity protective

Continued

Table 1 *Continued*

Category	Risk factor/modifier	Clinical insights and mechanisms
	Tobacco use	• Ever and current smokers have ∼1.2–1.5× increased PE risk • Risk elevated in cancer patients • Associated with endothelial dysfunction
Iatrogenic and transient	Surgery	• Orthopaedic procedures (hip/knee replacement) carry highest risk
	Trauma	• Early-onset PE common in patients admitted in ICU • Lower limb fractures and immobility key contributors • Mortality may reach 50% in some subgroups
Hereditary and haemostatic	factor V Leiden (FVL)	• Most common inherited thrombophilia (3–8× VTE risk) • Associated with reduced APC sensitivity
	Antiphospholipid syndrome	• Increases risk of arterial and venous thrombosis, including PE (3–10× higher risk) • Risk amplified during active disease flares or in the presence of lupus anticoagulant and triple antibody positivity
	Prothrombin G20210A mutation	• 2–3× increased risk • Elevated prothrombin levels lead to enhanced thrombin generation
	Natural anticoagulant deficiencies	• Antithrombin, Protein C, and Protein S deficiencies are rare but potent inherited VTE risk factors.
	Non-O blood type	• Non-O individuals have elevated vWF and FVIII levels • 1.5–2× VTE risk
	Elevated coagulation factors	• High levels of FVIII, FIX, FXI independently increase VTE risk • Often under-investigated
	Other genetic variants	• GWAS-identified genes (e.g. ABO, F11, KLKB1, SELP, SCUBE1) linked to thrombosis • Potential targets for future risk stratification
	Hyperhomocysteinemia	• Elevated homocysteine contributes to thrombosis • MTHFR polymorphism alone not consistently linked to PE

ABI, ankle-brachial index; APC, activated protein C; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; GWAS, genome-wide association studies; HF, heart failure; MASH, metabolic dysfunction-associated steatohepatitis (MASH); MTHFR, methylenetetrahydrofolate reductase; IBD, inflammatory bowel disease; ICU, intensive care unit; HR, hazard ratio; OR, odds ratio; PE, pulmonary embolism; RA, rheumatoid arthritis; SGLT2i, sodium glucose cotransporter 2 inhibitor; SLE, systemic lupus erythematosus; vWF, von Willebrand factor; VTE, venous thromboembolism.

per 100 000 US live births).[66] Acute PE during or after assisted reproductive technologies (ART), is predicted to rise due to the increased number of women undergoing these therapies. Data from the RIETE registry reported that 41 (0.6%) out of 6718 women of childbearing age with VTE had an ART-related event. Among these, 29.3% had isolated PE while 14.6% both PE and DVT.[67] While absolute risk remains low due to the limited number of ART-related VTE cases, elevated risks occur in those having unsuccessful ART cycles or experiencing ovarian hyperstimulation syndrome.[68]

Acute infectious illness

Acute infections, such as coronavirus disease-2019 (COVID-19), influenza, pneumonia, and urinary tract infections, transiently increase PE risk. Respiratory infections, including pneumonia, are major predisposing factors, with PE-related pulmonary infarction occurring in ∼one-third of cases.[69] Elevated PE risk may persist for months to a year after infection.[69] Evidence indicates that the increased risk of PE may persist beyond the acute phase of COVID-19.[70] A meta-analysis of 29 million patients reported that over a mean follow-up of 8.5 months, the cumulative incidence of PE in COVID-19 recovered patients was 1.2% (95% CI: 0.9–1.4, I2: 99.8%).[70] Furthermore, recovered COVID-19 patients presented a higher risk of incident PE (HR: 3.16, 95% CI: 2.63–3.79, I2 = 90.1%) compared to non-infected patients from the general population over the same follow-up period.[70]

Lifestyle-related risk and epidemiological trends

Obesity

Overweight and obesity are key modifiable risk factors for PE.[71] In the United States, as of 2018, approximately 30% of adults were classified as overweight and 42.4% as obese, with severe obesity affecting 9.2% of the population.[72] More recent data from 2021 to 2023 suggest a slight stabilization in overall obesity rates, with 40.3% of adults now classified as obese.[73] However, this remains well above the Healthy People 2030 target of 36%.[74] Severe obesity continues to rise, particularly among adults aged 40–59 years, where prevalence has reached 46.4%.[75] In Europe, obesity prevalence varies widely by country but impacts roughly 23% of adults, with higher rates in Southern and Eastern Europe compared to Northern countries.[76] Obesity significantly influences PE epidemiology, conferring a 2- to 2.5-fold increased risk of first-time PE and contributing substantially to the rising population burden of PE, particularly in regions with high prevalence of severe obesity such as the United States.[71]

A comprehensive meta-analysis involving over 3 910 747 participants found that obesity is associated with a 2.24-fold increased risk of PE compared with individuals with a lean body mass index.[77] Despite being considered a moderate risk factor relative to other predisposing factors for VTE and PE,[78] obesity's growing prevalence in the population makes it a critical public health concern.[79] Obesity interacts with other risk factors, such as immobility and inflammation,[80] amplifying the likelihood of both first (OR, 2.1; 95% CI, 2.0–2.3) and recurrent VTE events (HR: 1.3; 95% CI, 0.9–1.9).[81,82] The pathophysiological mechanisms underlying this increased risk include venous stasis, chronic inflammation, and procoagulant changes associated with excess adipose tissue.[83]

While obesity increases the risk of developing PE, it paradoxically appears to be associated with more favourable survival in patients with PE, a phenomenon known as the obesity paradox. Obesity is included in most risk assessment tools for VTE prophylaxis, such as the Padua Prediction Score[84] and the IMPROVE score[85] which guide clinicians in implementing preventive measures for at-risk hospitalized patients. However, obesity is currently not included in prognostic tools for PE, like the Pulmonary Embolism Severity Score[86] and its simplified version.[87] Recognizing obesity as a modifiable risk factor emphasizes the importance of weight management strategies in reducing the incidence of PE and related recurrence.[71]

Sedentary lifestyle

A sedentary lifestyle is a significant risk factor for acute PE as highlighted by various studies demonstrating a strong association between physical inactivity and the incidence of idiopathic PE.[88,89] In a large prospective cohort of 69 950 female nurses, spending >40 h per week in sedentary was associated with a greater than two-fold increased risk of PE (hazard ratio, 2.34) compared with less sedentary participants.[88] Conversely, a recent meta-analysis showed that greater levels of physical activity are associated with reduced VTE risk. In contrast, every additional hour of sedentary time per day is associated with a 2% greater risk of VTE.[89] Additionally, prolonged immobility during long-distance travel, including flights or extended bus/train journeys exceeding 4–6 h, represents a commonly-identified minor transient risk factor for PE, particularly in individuals with additional predisposing conditions such as obesity, prior VTE, cancer, or thrombophilia.[90]

Smoking

While less well-studied than other cardiovascular outcomes, smoking modestly increases PE risk.[91] Meta-analysis data show relative risks of 1.17 for ever smokers, 1.23 for current smokers, and 1.10 for former smokers.[92] Only 30%–40% of patients quit after serious cardiovascular events, despite continued smoking compounding the risk of subsequent arterial thrombotic events.[92] Definitive data regarding a potential association between electronic cigarette use or vaping and the risk of acute PE are lacking.[93]

Drug-associated risk factors

In addition to classical risk factors, several medications have been associated with increased risk of VTE and PE. Oral contraceptives and hormone replacement therapy both increase VTE and PE risk.[94] Case–control studies demonstrate that combined oral contraceptives (COCs) containing the third-generation progestogens in combination with ethinyl oestradiol increase the risk of VTE 1.5- to 3-fold compared to oral contraceptives containing the second-generation progestogen levonorgestrel plus oestrogens.[95] Of note, VTE risk is specifically influenced by factors such as oestrogen dosage and type of progestin used. For instance, OCs with higher doses of oestrogen (>50 μg) are associated with a higher risk of VTE compared to lower-dose formulations. Norpregnane derivatives, a class of progestins present in both combined oral contraceptives and hormone replacement therapy as well as fourth-generation oral contraceptives containing drospirenone or dienogest are associated with an approximately four-fold increased risk of VTE and PE.[96] This risk of VTE is further influenced by factors such as age, smoking, obesity, first degree family member history and genetic mutations such as factor V Leiden and prothrombin gene mutation and other individual predispositions to thrombosis.[65]

Hormone replacement therapy used to treat symptoms of menopause and perimenopause is associated with a two- to four-fold increased risk of VTE compared to non-users. Oral oestrogen therapy elevates the risk for VTE, with an odds ratio (OR) of 4.2 (95% CI, 1.5–11.6) in current users compared with non-users, while transdermal oestrogen has a much lower odds ratio (OR, 0.9, 95% CI, 0.4–2.1).[65] The odds of VTE are highest during the first year of treatment, with an OR of 4.0, compared with 2.1 for treatment beyond one year. In combined oral contraceptive users, the route of administration (oral vs transdermal) does not appear to attenuate VTE risk, whereas in hormone replacement therapy users, transdermal oestrogen appears to be safer than oral formulation, being associated with a lower thromboembolic risk.[94] Vaginal oestrogen similarly does not appear to elevate VTE risk.[94,97]

Testosterone therapy, increasingly prescribed for hypogonadism and age-related androgen deficiency, has been linked to an elevated short-term risk of VTE, particularly within the first months of treatment.[98] Proposed mechanisms include increased haematocrit, enhanced platelet aggregation, and procoagulant changes. The risk appears higher in individuals with underlying thrombophilia or other predisposing factors.[98]

Erythropoiesis-stimulating agents, frequently used in patients with cancer or chronic kidney disease (CKD), are associated with increased thromboembolic risk, particularly when targeting higher haemoglobin levels.[99] The prothrombotic effect is thought to relate to increased blood viscosity, endothelial activation, and interaction with underlying inflammatory and malignant states.[99]

Finally, antipsychotic medications, especially second-generation agents, have also been associated with increased VTE risk.[100] Proposed mechanisms include sedation-related immobility, weight gain, cardiometabolic syndrome, hyperprolactinemia, and possible direct procoagulant effects. The risk appears greatest during the early phase of treatment and in patients with additional predisposing conditions.[100]

Prior venous thromboembolism

Prior VTE represents one of the strongest predictors of recurrent PE and is a major enduring risk factor. Patients with a history of VTE have a substantially elevated risk of recurrence, particularly during the first year after discontinuation of anticoagulation, but with a persistent long-term excess risk thereafter.[1] Recurrence risk varies according to the nature of the initial event: it is significantly higher following unprovoked VTE compared with events associated with major transient risk factors such as surgery or trauma.[3,40] The presence of persistent prothrombotic conditions (e.g. active cancer, chronic inflammatory disease, or inherited thrombophilia) further amplifies recurrence risk.[101] Importantly, recurrent events frequently present as PE, even when the index event was DVT, underscoring the clinical relevance of prior VTE as a determinant of future PE.[14]

Thrombophilia

Inherited and acquired thrombophilias represent important enduring risk factors for VTE and PE. Common inherited conditions include Factor V Leiden mutation, which is found in approximately 5%–8% of individuals of European ancestry and confers a three- to seven-fold increased risk of VTE, and prothrombin G20210A mutation, noted in 2%–4% of Europeans and contributing to a two- to three-fold risk increase.[102] Deficiencies of natural anticoagulants such as antithrombin, protein C, and protein S are rarer (<1%) but associated with markedly higher thrombotic risk, depending on the extent of deficiency.[102] Acquired thrombophilias, particularly antiphospholipid antibody syndrome, impact up to 5% of patients with unprovoked VTE and substantially amplify incident PE susceptibility and recurrence risk, especially when combined with transient risk factors like surgery, pregnancy, or malignancy.[103] Overall, thrombophilia not only predisposes patients to first-time events but also increases the likelihood of recurrence, with approximately 30%–40% of individuals experiencing recurrent VTE over 10 years in the presence of persistent prothrombotic conditions.[102] Recognition of thrombophilia is therefore critical for individualized risk stratification, guiding both preventive strategies in high-risk settings and long-term anticoagulation management, particularly in patients with persistent or combined risk factors.[1]

Major transient risk factors

Surgery

Surgery increases the risk of acute PE by triggering a hypercoagulable and proinflammatory state, causing endothelial injury, and venous stasis from immobility, all of which promote thrombus formation.[14] Elective orthopaedic surgeries, particularly total hip replacement and total knee replacement, are well-established risk factors for VTE, including PE. Previous investigations have reported that the incidence of symptomatic PE after total hip replacement, within the first 90 days, is around 0.46%–0.68%.[104] Similarly, the incidence of symptomatic PE after total knee replacement has been reported between 0.27% and 0.6% in the 3 months after the procedure.[104]

Trauma

The incidence of PE among trauma patients varies widely, ranging from 0.35–24%, with studies indicating that this risk is particularly pronounced in patients admitted to the intensive care unit.[105] In one study, 37% of PE events occurred within the first 4 days after injury.[106] Factors such as fractures of the lower extremities, obesity, and age are frequently associated with PE, especially early-onset PE.[107] Trauma patients undergoing major orthopaedic surgery like hip fracture repair have increased risk of PE with incidence ranging from 0.7% to 30% in the absence of adequate thromboprophylaxis.[108] Fatality rates from PE approach 50% in some series, underscoring the severity of this complication.[107]

Chronic comorbidities in PE: impact on prognosis

Beyond shared risk factors, there are multiple associations between acute PE and many chronic cardiovascular and non-cardiovascular comorbidities, resulting in an interdependence in the risk of the disease and its prognosis (*Figure 2*).

Cardiovascular comorbidities

Heart failure

HF is independently associated with a heightened risk of early complications in patients with acute PE, including a mortality rate nearly three times higher than that of non-HF patients (11.27% vs 4.33%,) and double the incidence of major bleeding events (9.86% vs 4.54%).[109] Current prognostic tools for acute PE may inadequately stratify risk in patients with concomitant history of HF, as traditional markers like right ventricular-to-left ventricular diameter ratios show reduced predictive value in those with pre-existing cardiac remodelling and increased levels of brain natriuretic peptides.[110] In recent years, pharmacologic advances in HF management have expanded to include to include sodium-glucose co-transporter 2 inhibitors (SGLT2i)[111] and GLP-1 receptor agonists and dual GIP/GLP-1 receptor agonists such as tirzepatide.[112] By promoting significant improvement on the natural history of HF and cardiometabolic risk, these agents may indirectly mitigate susceptibility to PE. Future studies are needed to formally evaluate the impact of these agents on VTE and PE outcomes in HF cohorts, especially in high-risk subgroups.

Atherothrombosis

Patients with acute PE and pre-existing coronary artery disease (CAD) face a poorer prognosis, with epidemiological studies demonstrating strong reciprocal associations. Patients with a VTE event exhibit a 20% increased risk for future arterial ischaemic events.[113] A population-based case-control study found that recent hospitalization for acute myocardial infarction increased the risk of isolated PE six-fold (OR 6.3; 95% CI 5.5–7.2).[114] Furthermore, long-term follow-up data from a cohort of 1023 PE patients demonstrated that a previous history of cardiovascular diseases, including CAD, was an independent predictor of post-discharge mortality, accounting for 36% of deaths during median 4.7-year follow-up.[115] Therefore, the interplay between CAD and PE extends beyond acute events, due to the sharing of pathophysiological mechanisms like endothelial dysfunction and hypercoagulability.[116]

PE complicates ∼0.2%–0.8% of stroke cases, with higher incidence in severe strokes, cancer, or prior VTE. These patients face higher 30-day (25.8% vs 13.6%) and 1-year mortality (47.2% vs 24.6%), increased disability, and longer hospitalizations.[117] The risk factors for PE in stroke patients mirror those

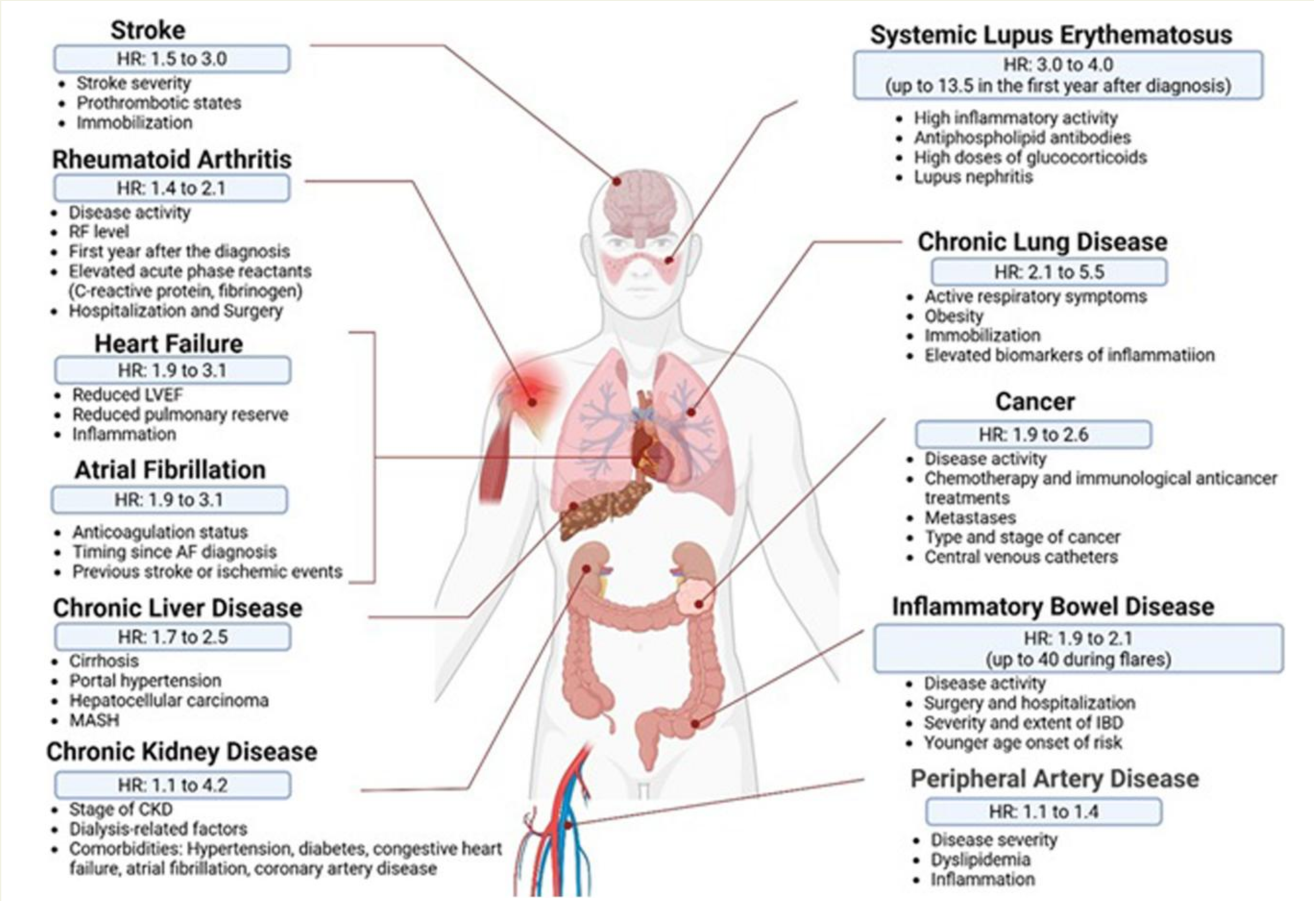

Figure 2 Association between risk of acute pulmonary embolism and common chronic cardiovascular and non-cardiovascular conditions. The main factors that increase the risk of acute pulmonary embolism are reported below. All risks are expressed as hazard ratios to enable comparability across different diseases. The values shown represent the minimum and maximum hazard ratios identified from the literature search. AF, atrial fibrillation; CKD, chronic kidney disease; HR, hazard ratio; IBD, inflammatory bowel disease; LVEF, left ventricular ejection fraction; NASH, non-alcoholic steatohepatitis; RF, rheumatoid factor

in the general population, including age, immobility, and thrombophilia.[118] Prognostically, PE in patients with ischaemic stroke is associated with a higher risk of adverse outcomes.[119] The concurrent management of both conditions is challenging due to the need for anticoagulation in PE and thrombolysis in ischaemic stroke, which poses higher risk of intracranial bleeding.[117]

Peripheral artery disease (PAD) is associated with an increased risk of acute PE (HR: 1.12; 95% CI: 1.06–1.18). Furthermore, the risk VTE in patients with PAD has been associated with the severity of the disease.[120] Patients with severely low ABI (0.00–0.39) or elevated ABI (>1.40) have the highest risk of acute PE (HR: 1.46; 95% CI: 1.31–1.64) compared with patients with normal ABI.[120] Polyvascular disease, defined as atherosclerotic involvement of multiple vascular beds, is associated with a graded increase in VTE risk, including a 62% greater risk with two affected territories compared with one (HR; 1.62, 95% CI 1.09–2.40) and more than a two-fold risk with three territories (HR: 2.04, 95% CI 1.10–3.77).[121]

Atrial fibrillation

Atrial fibrillation (AF) demonstrates a complex temporal relationship with PE, serving as both a risk factor, sequela, and prognostic marker across disease stages. Epidemiological data reveal AF prevalence of 18.1% (18 126 per 100 000) in acute PE populations, with a further 12.6% risk of developing new-onset AF during follow-up.[122] Registry studies have demonstrated bidirectional associations. Specifically, a Danish analysis shows PE patients have 3.3-fold higher AF prevalence than matched controls,[123] while Taiwanese cohorts identify AF as an independent PE risk factor (HR 1.72; 95% CI 1.10–2.71).[124] The thromboembolic risk escalates dramatically post-AF diagnosis, with 10.88-fold increased PE incidence (95% CI 6.23–18.89) and 6.16-fold ischaemic stroke risk (95% CI 4.47–8.48) within the first six months.[123] In acute PE, the presence of concomitant AF significantly worsens outcomes. In fact, patients with pre-existing AF exhibit 59% mortality vs 35% in non-AF counterparts during long-term follow-up, while incident AF during hospitalization predicts a 77% increase in-hospital mortality.[125]

Non-cardiovascular comorbidities

Chronic lung disease

Acute PE is relatively common in chronic lung disease, including COPD and interstitial lung disease.[126,127] Reported PE prevalence in COPD exacerbations varies widely (3.3%–25%) depending on setting and diagnostic approach, with multicentre studies estimating ∼5.9% and meta-analyses 3.6%–19.9%.[128,129] COPD patients with PE suffer worse outcomes, including increased short-term mortality and longer hospital stays (OR 2.84; 95%CI 2.27–3.55).[126] Both stable COPD and acute exacerbations may increase the risk of VTE.[129] Moreover, the clinical presentation represents an additional challenge, as patients with PE and COPD often exhibit overlapping symptoms, such as dyspnoea and chest tightness, which can lead to delayed diagnosis.[126]

Several investigations have also shown that asthma is associated with an increased risk of PE. A nationwide cohort study reported that the adjusted hazard ratio for PE in asthmatic patients was 3.24 compared with non-asthmatic individuals.[130] Furthermore, a case-control study from Sweden reported an odds ratio of 1.43 for PE and of 1.60 for combined PE and DVT in asthma patients.[131] Moreover, the incidence of PE in patients with severe asthma was reported as 0.93 per 1000 person-years, higher than in those with mild-to-moderate asthma (0.33 per 1000 person-years) or the general population (0.18 per 1000 person-years).[132]

Chronic kidney disease

CKD is associated with a complex number of haemostatic alterations, including both procoagulant and anticoagulant effects, which can influence the risk of VTE as well as bleeding.[133] The estimated annual frequency of PE was 527 per 100 000, 204 per 100 000, and 66 per 100 000 population with ESRD, CKD, and normal renal function, respectively.[134] The Longitudinal Investigation of Thromboembolism Etiology study, a large population-based investigation, found that CKD increases the risk of VTE.[135] Specifically, the incidence rates per 1000 person-years were 1.5 for normal kidney function, 1.9 for mildly decreased renal function, and 4.5 for stage 3 or 4 CKD.[135] The relative risk of VTE with stage 3 or 4 CKD was 2.1 (95% CI 1.5–3.0).[135]

Cancer

Acute PE represents a significant, and potentially life-threatening, complication in patients with cancer and thrombotic risk varies substantially by cancer type, stage, and treatment.[136] Certain malignancies, including pancreatic, gastric, lung, brain, biliary, and gynaecological cancers, are consistently associated with the highest PE incidence, ranging from 10% to 35%, whereas breast and prostate cancers confer comparatively lower risk.[52] Patients with active metastatic disease exhibit markedly elevated risk, in part due to tumour-associated procoagulant activity, including production of tissue factor-laden microparticles, systemic inflammation, and immobility.[2] Chemotherapy, particularly cisplatin-based regimens, hormonal therapies, and some targeted agents, further amplify thrombotic susceptibility.[52]

Incidental or unsuspected PE is common in cancer patients, detected in 1%–5% of CT scans performed for other indications, and may account for approximately half of all PE diagnoses in oncology.[137] Despite often being asymptomatic, incidental PE appears to confer similar adverse outcomes as symptomatic events, supporting the recommendation for anticoagulation management in most cases.[138]

The presence of metastasis, poorer performance status, and certain cancer-directed therapies further increase the likelihood of recurrent VTE, especially when the initial event is a PE (OR = 10.5 95% CI: 9.3–11.7).[139] Individualized risk stratification using cancer type, disease stage, performance status, and ongoing therapy has been advocated to guide both prophylactic and therapeutic anticoagulation, balancing thrombotic and bleeding risks.[140] Subsegmental PE, which is common in cancer patients, may have different outcomes compared to non-cancer populations, supporting the recommendation for anticoagulation treatment even in incidentally-detected cases.[141] These distinctions underscore that cancer is not a uniform VTE risk factor but rather represents a spectrum of thrombotic susceptibility. Accordingly, such observations highlight the importance of personalized management strategies in oncology patients.

Inflammatory chronic diseases

Chronic inflammatory diseases, including rheumatoid arthritis, systemic lupus erythematosus, and inflammatory bowel disease, increase acute PE risk. Exacerbations of such inflammatory disorders may function as a transient risk factor while chronic disease represents an enduring risk. In patients with RA, the risk of acute PE has been reported as 2.16 (95% CI: 1.75–2.86), after adjusting for concomitant comorbidities.[142] Similarly, a large meta-analysis, based on more than 250 000 patients with rheumatoid arthritis, confirmed a higher risk of acute PE in these patients (OR: 2.15; % CI: 1.39–3.49) compared with the general population.[143] In inflammatory bowel disease the incidence rate of acute PE is 10.3 per 10 000 person-years and 19.8 per 10 000 person-years in patient with Crohn's Disease and ulcerative colitis, respectively.[144] In such patients, the risk of acute PE is higher among hospitalized patients (OR 1.85 for ulcerative colitis and OR 1.48 for Crohn's disease, respectively),[145] during acute phases and flares[146] and two- to three-fold higher with the use of glucocorticoids.[147] The annual incidence of acute PE in hospitalized patients with systemic lupus erythematosus ranges between 1.2% and 1.6%.[148] Incidence rates for PE in patient with SLE was estimated to be 2.58 per 1000 person-years.[149] High disease activity and a disease duration <1.5 years is associated with an increased risk of acute PE (OR: 3.85; 95% CI: 1.82–7.97).[148]

Chronic liver disease

Chronic liver disease, including non-cirrhotic conditions, is associated with an increased risk of acute PE.[150] The incidence of PE, ranges from 0.2% to 6.3% depending on the study and population characteristics.[151] A large meta-analysis based on 11 retrospective studies, including 695 012 cirrhotic patients and 1 494 660 non-cirrhotic controls reported a higher risk of acute PE in these patients (OR = 1.65; 95% CI 1.04–2.63).[152] Moreover, a retrospective case-control study highlighted a greater prevalence (79.2% vs 53.3%) and risk (HR: 4.33) of acute PE in patients with metabolic dysfunction-associated steatohepatitis compared to controls.[153]

Looking forward: precision management and prevention to modify the PE epidemiology

PE prevention is moving beyond anticoagulation towards precision medicine. Therapies like GLP-1 and dual GIP/GLP-1

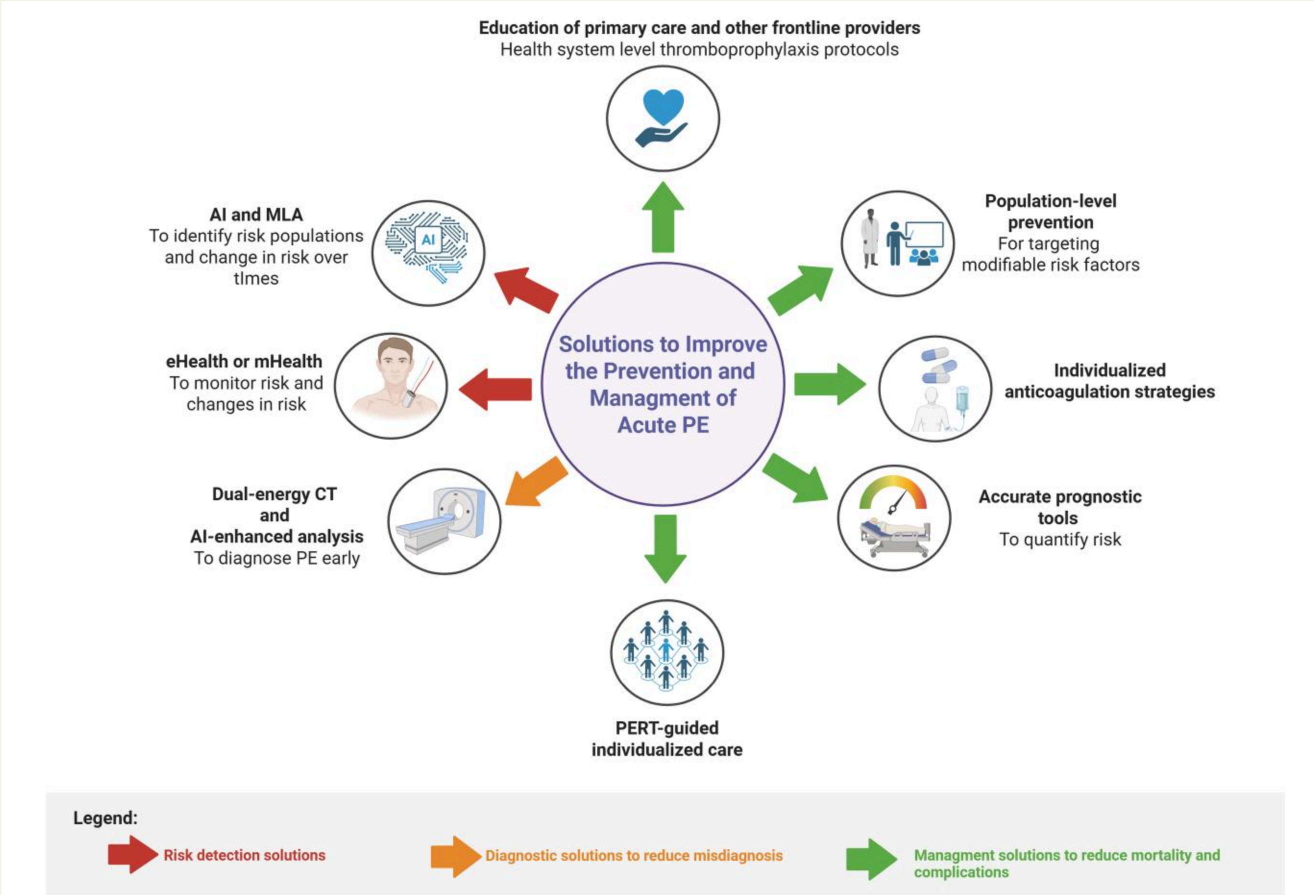

Figure 3 Potential solutions to improve the prevention and management of acute pulmonary embolism to reduce its epidemiological burden. AI, artificial intelligence; CT, computed tomography; MLA, machine learning algorithm; mHealth, mobile health; PE, pulmonary embolism; PERT, pulmonary embolism response team

agonists, which reduce weight, visceral adiposity, insulin resistance, and inflammation, each a thromboembolic risk factor, may lower PE risk in select population.[154] Looking ahead, clinical evaluation and management pathways for acute PE are increasingly being shaped by advances in personalized medicine and targeted prevention strategies, which are expected to reduce incidence rates, improve risk stratification, and optimize outcomes in high-risk populations (*Figure 3*).[155]

While technological innovation and individualized management are reshaping the diagnostic and therapeutic landscape of acute PE, meaningful reductions in incidence and mortality will also depend on population-level prevention strategies. Primary prevention targeting modifiable risk factors, such as obesity, sedentary behaviour, smoking, and inadequate thromboprophylaxis during hospitalization, remains foundational. Public health initiatives aimed at increasing awareness of early PE symptoms (e.g. unexplained dyspnoea, pleuritic chest pain, syncope) and promoting timely medical evaluation may facilitate earlier diagnosis and reduce fatal presentations.[1] Moreover, systematic implementation of evidence-based thromboprophylaxis protocols in medical and surgical settings, along with education of primary care providers and high-risk patients, represents a high-impact, scalable strategy. Integrating these public health measures with precision diagnostics and advanced therapeutics offers a more comprehensive framework to modify PE epidemiology at both individual and population levels. Integrating multi-omics data, such as genomics, epigenomics, transcriptomics, proteomics, metabolomics, and microbiomics, may uncover novel molecular pathways and biomarkers for improved risk stratification and targeted therapies.[156] Furthermore, artificial intelligence (AI) and machine learning algorithms, trained on representative datasets incorporating clinical, imaging, and multi-omics data, may offer unprecedented capabilities for enhancing diagnostic accuracy, refining risk prediction models, and defining clinically relevant PE phenotypes.[157]

An essential component of precision management and prevention in PE is ensuring adherence to validated diagnostic and therapeutic strategies. Recent evidence underscores that system-level interventions promoting adherence to both diagnostics and anticoagulation remain critical to achieving meaningful improvements in patient care.[158] Optimized use of guideline-recommended diagnostic algorithms, including clinical risk scores (e.g. PESI/sPESI) informing pre-test probability-based imaging pathways, facilitates timely and accurate identification of PE, particularly in atypical or high-risk presentations. Equally important is strict adherence to evidence-based anticoagulation, including appropriate agent selection, dosing, and duration, tailored to patient-specific factors, such as age, renal function, comorbidities, cancer status, and presence of thrombophilia.

Enhanced epidemiologic understanding complements, such approaches, informing targeted interventions, guiding clinical trial populations, and addressing social determinants of health to reduce disparities. Together, these strategies aim to decrease PE incidence, optimize outcomes, and refine management pathways in high-risk populations.[14]

Conclusion

The relationship between understanding of PE epidemiology and preventive and therapeutic interventions is at a pivotal juncture. The rising global burden of this potentially life-threatening condition and growing population at risk demands a paradigm shift towards proactive and personalized preventive measures driven by a greater understanding of the disease epidemiology.

Supplementary data

Supplementary data are available at the *European Heart Journal* online.

Declarations

Disclosure of Interest

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Data Availability

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